2D shapes and 3D solids — Area · Surface area · Volume · Units

Remember the units in every answer: cm² for area, cm³ for volume. A calculator is allowed for questions marked □. □ = challenge question.

A · AREA OF 2D SHAPES

1 Work out the area of

- **a** a triangle with base 6 cm and perpendicular height 4.5 cm
- **b** a parallelogram with base 9 cm and perpendicular height 5 cm
- **c** a trapezium with parallel sides 8 cm and 14 cm, and height 5 cm.

> **a** $\frac{1}{2} \times 6 \times 4.5 = 13.5$ cm². **b** $9 \times 5 = 45$ cm². **c** $\frac{1}{2}(8 + 14) \times 5 = 11 \times 5 = 55$ cm².

2 A shape is made of a rectangle 12 cm wide and 6 cm tall, with a triangle of height 4 cm sitting on its full width. Work out the total area.

> Rectangle $12 \times 6 = 72$ cm²; triangle $\frac{1}{2} \times 12 \times 4 = 24$ cm². Total 96 cm².

3 Calculate the shaded area:

- **a** a square of side 10 cm with a square hole of side 3 cm cut from it
- **b** a rectangle 12 cm by 8 cm with a triangle (base 12 cm, height 8 cm) removed.

> **a** $100 - 9 = 91$ cm². **b** $96 - \frac{1}{2} \times 12 \times 8 = 96 - 48 = 48$ cm² — exactly half.

4 Two triangles have the same area. The first has base 8 cm and height 6 cm. The second has base 12 cm. Work out its height.

> Area $= \frac{1}{2} \times 8 \times 6 = 24$ cm². For the second: $\frac{1}{2} \times 12 \times h = 24 \Rightarrow 6h = 24 \Rightarrow h = 4$ cm.

5 A parallelogram has base 6 cm and perpendicular height 35 mm. Work out its area in cm².

> Same units first: 35 mm $= 3.5$ cm. Area $= 6 \times 3.5 = 21$ cm².

B · 3D SOLIDS

6 A cuboid measures 3 cm by 4 cm by 7 cm.

a Work out its volume.

b Work out its surface area.

c It stands on its 7 cm by 4 cm face. Give the dimensions of its plan view, front elevation and side elevation.

> **a** $3 \times 4 \times 7 = 84$ cm³.
> **b** Three pairs of faces: $2(3 \times 4) + 2(3 \times 7) + 2(4 \times 7) = 24 + 42 + 56 = 122$ cm².
> **c** Plan (from above): 7 by 4. Front elevation: 7 by 3. Side elevation: 4 by 3.

7 An L-shaped prism is 8 cm long. Its cross-section is a 10 cm by 2 cm rectangle with a 3 cm by 4 cm rectangle attached. Work out the volume of the prism.

> Area of cross-section $= 10 \times 2 + 3 \times 4 = 32$ cm². Volume = cross-section × length $= 32 \times 8 = 256$ cm³.

8 An open box (no lid) is a cuboid 20 cm long, 15 cm wide and 10 cm high. Work out the area of card needed to make it.

> Base $20 \times 15 = 300$; two long sides $2 \times (20 \times 10) = 400$; two short sides $2 \times (15 \times 10) = 300$. Total $1\,000$ cm² (no top).

9 A cuboid has volume 480 cm³, length 10 cm and height 6 cm. Work out its width.

> $10 \times w \times 6 = 480 \Rightarrow 60w = 480 \Rightarrow w = 8$ cm.

10 Write down the number of faces, edges and vertices of **a** a triangular prism, **b** a square-based pyramid.

> **a** 5 faces (2 triangles, 3 rectangles), 9 edges, 6 vertices. **b** 5 faces (1 square, 4 triangles), 8 edges, 5 vertices.

C · UNITS AND PROBLEMS

11 Copy and complete:

a $2.5\ \text{m}^2 = \square\ \text{cm}^2$ **b** $7\,500\ \text{mm}^3 = \square\ \text{cm}^3$ **c** $0.8\ \text{litres} = \square\ \text{cm}^3$

d $3\ \text{cm}^2 = \square\ \text{mm}^2$

> $1\ \text{m}^2 = 100 \times 100 = 10\,000\ \text{cm}^2$; $1\ \text{cm}^3 = 10 \times 10 \times 10 = 1\,000\ \text{mm}^3$; 1 litre $= 1\,000$ cm³; $1\ \text{cm}^2 = 100\ \text{mm}^2$.
> **a** $25\,000$ **b** 7.5 **c** 800 **d** 300.

12 □ Small boxes measure 10 cm by 5 cm by 4 cm. They are packed into a crate 50 cm by 30 cm by 25 cm, with the 10 cm sides along the 50 cm length, the 5 cm sides along the 30 cm width and the 4 cm sides upright.

a How many boxes fit in the crate?

b What volume of the crate is left empty?

> **a** Along the length $50 \div 10 = 5$; across $30 \div 5 = 6$; layers $25 \div 4 = 6$ (remainder 1 cm). $5 \times 6 \times 6 = 180$ boxes.
>
> **b** Crate $50 \times 30 \times 25 = 37\,500$ cm³; boxes $180 \times 200 = 36\,000$ cm³; empty $1\,500$ cm³ (the 1 cm gap at the top).

13 A trapezium has parallel sides of 9 cm and 15 cm and an area of 72 cm². Work out its height.

> $\frac{1}{2}(9 + 15) \times h = 72 \Rightarrow 12h = 72 \Rightarrow h = 6$ cm.

Formulas: triangle $\frac{1}{2}bh$ · parallelogram bh · trapezium $\frac{1}{2}(a + b)h$ · cuboid lwh · prism = cross-section × length · surface area = add every face.